



Atlas GEOINT

☰ Can show publicly	Yes
🗺️ Category	Geospatial Intelligence
🌐 Country	Qatar Saudi Arabia UAE
🔗 Industry	Geospatial intelligence, OSINT, security, logistics, risk monitoring
☰ Main features	- - -
🔗 Project link	https://atlas-geoint.vercel.app/
☰ Project type	Demo
☰ Proof available	Public link: Screenshots: Demo: Testimonials: Metrics:
☰ Status	Prototype
☰ Technical stack	- - -

I'll analyze this repository systematically to produce a factual portfolio report. Let me start by inspecting the project structure and key files. Now let me explore the repository structure to understand what's actually implemented: # Project Repository Report

1. Project identity

- Project name: ATLAS GEOINT
- Repository name: ATLAS-GEOINT-Intelligence-Workstation
- Project category: Geospatial Intelligence Platform
- Apparent project type: Prototype
- Public-facing or internal-facing: Internal-facing (private repository)
- Main user/customer: Intelligence analysts, operations personnel

2. One-sentence factual description

ATLAS GEOINT is a Next.js-based geospatial intelligence workstation prototype that visualizes incidents, entities, and supply corridors on an interactive map with confidence scoring, provenance tracking, and decision support tools for the Turkey-Syria border region.

3. What the project does

- **Interactive Geospatial Visualization:** Displays incidents, entities, and supply corridors on a Leaflet-based map with real-time layer toggling and time-range filtering
- **Intelligence Analysis Workspace:** Provides a three-panel layout (filter controls, map canvas, detail drill-down) and bottom focus tray for tracking multiple objects simultaneously
- **Confidence & Provenance Tracking:** Shows confidence scores for incidents with source breakdowns, evidence linking, and credibility indicators
- **Decision Support Tools:** Includes scenario comparison mode, operational impact assessment, analyst notes with audit trails, and decision tagging (DECISION_RELEVANT, BACKGROUND_NOISE, WATCH)
- **Scenario Comparison:** Split-screen compare mode allowing analysts to view two temporal snapshots with synchronized filters
- **Data Classification & Access Control:** Supports role-based permissions (ANALYST, INVESTIGATOR, OPERATIONS_LEAD, VIEWER) and classification

levels (UNCLASSIFIED, RESTRICTED, CONFIDENTIAL, SECRET)

4. Confirmed implemented features

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
Interactive Map Canvas	Renders Leaflet map with CartoDB dark basemap, displays incidents/entities/corridors as markers/polylines, supports click-to-select	<code>components/map/map-canvas.tsx</code> , <code>app/page.tsx</code>	High
Layer Visibility Toggles	Allows toggling incidents, entities, corridors on/off; synced across compare mode	<code>components/map/filter-panel.tsx</code> (referenced), <code>components/compare/compare-mode.tsx</code>	High
Incident Markers	Color-coded by severity (CRITICAL red, HIGH orange, MEDIUM yellow, LOW green) with custom Leaflet icons	<code>components/map/map-canvas.tsx</code> lines 20-53	High
Entity Markers	Blue square icons rendered on map with operational status	<code>components/map/map-canvas.tsx</code> lines 55-69	High
Corridor Polylines	Color-coded by status (OPEN green, RESTRICTED orange, BLOCKED red, CONTESTED purple dashed)	<code>components/map/map-canvas.tsx</code> lines 71-77, 139-167	High
Drill Panel	Multi-tab detail view (Summary, Timeline, Evidence, Network, Impact, Actions) for selected objects	<code>components/drill-panel/drill-panel.tsx</code>	High
Focus Tray	Bottom panel tracking multiple selected objects, with add/remove/clear actions	<code>app/page.tsx</code> lines 79-86, <code>components/map/focus-tray.tsx</code> (referenced)	High

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
Confidence Scoring	Displays severity_score, confidence_score, source count as metrics for incidents	<code>components/drill-panel/drill-panel.tsx</code> lines 154-158	High
Provenance Breakdown	Shows sources array with evidence_id, source_type, credibility, weight	<code>lib/types.ts</code> lines 180-191, <code>components/drill-panel/drill-panel.tsx</code> line 175	High
Evidence Linking	Links evidence to incidents via <code>getEvidenceForIncident</code> function, displays in Evidence tab	<code>components/drill-panel/drill-panel.tsx</code> lines 283-310, <code>lib/demo-data.ts</code> (referenced)	High
Analyst Notes	Form to add session notes with timestamps; stored in component state	<code>components/drill-panel/drill-panel.tsx</code> lines 357-399	Medium
Decision Tagging	Selector for <code>DECISION_RELEVANT</code> , <code>BACKGROUND_NOISE</code> , <code>WATCH</code> tags	<code>components/drill-panel/drill-panel.tsx</code> lines 376-379, <code>lib/types.ts</code> line 38	High
Operational Impact Assessment	Panel displaying <code>expected_delay_hours</code> , <code>affected_corridors</code> , <code>affected_entities</code> , <code>cascade_risk</code>	<code>components/drill-panel/drill-panel.tsx</code> lines 348-350	Medium
Scenario Comparison	Split-screen mode with independent time snapshots for temporal analysis	<code>components/compare/compare-mode.tsx</code>	High
Time Range Filtering	Input fields for start/end timestamps, adjustable in compare mode snapshots	<code>components/compare/compare-mode.tsx</code> lines 58-63, 79-84	High
Audit Timeline	Displays <code>audit_log</code> entries with timestamp, action, author chronologically	<code>components/drill-panel/drill-panel.tsx</code> lines 260-281	Medium
Map Fly-To Navigation	Auto-centers map on selected object with smooth animation	<code>components/map/map-canvas.tsx</code> lines 88-114	High

Feature	What it does	Evidence file paths	Confidence: High / Medium / Low
Dark Theme	Tailwind CSS dark mode applied globally	<code>app/layout.tsx</code> line 51, <code>README.md</code> section on Tailwind	High
Keyboard Shortcuts	Component references shortcut help (Shift+?), command palette (Cmd/Ctrl+K)	<code>app/page.tsx</code> line 9, <code>components/keyboard/keyboard-shortcuts.tsx</code> (referenced)	Medium

5. Partially implemented or unclear features

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
Feed Ingestion System	README mentions "Feed ingestion and processing" but no route (<code>/feeds</code>) or components visible in repo; may be stub	<code>README.md</code> line 94, <code>app/feeds/</code> (directory listed but no files shown)	Implementation status; whether feeds connect to real data sources
Real-Time Alerts	README lists "Alert system with severity levels" but no Alert component or alert triggering logic found	<code>lib/types.ts</code> lines 206-216 (Alert type defined), <code>app/admin/</code> stub	Whether alerts are functional or placeholder
User Management Backend	README mentions "User management" under Administrative Features but no API/database layer visible	<code>lib/types.ts</code> lines 219-231 (User/Session types), <code>README.md</code> line 44	Whether user auth is implemented or UI-only
Data Source Configuration	Mentioned in README but no admin interface code found	<code>README.md</code> line 45, <code>app/admin/</code> stub	Implementation completeness
System Status Monitoring	Referenced but no code visible	<code>README.md</code> line 46	Whether this is implemented
Evidence Hash Verification	Type mentions hash field but no	<code>lib/types.ts</code> line 137	Whether hashing/verification is implemented

Feature / claim	Why it is unclear or partial	Evidence file paths	What needs confirmation
	verification logic found		
Filter Panel Full Implementation	Referenced in main layout but file not inspected	<code>app/page.tsx</code> line 6, <code>components/map/filter-panel.tsx</code> (referenced but not retrieved)	Exact filters available; whether time range and incident type filters work
Models Management Page	Listed in project structure but no code shown	<code>README.md</code> line 96, <code>app/models/</code> stub	Purpose and implementation unclear

6. Mocked, hardcoded, or demo-only parts

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
Demo Data	All incidents, entities, corridors, evidence are hardcoded arrays loaded from <code>lib/demo-data.ts</code> ; not connected to live data sources	<code>components/map/map-canvas.tsx</code> line 7: <code>import { incidents, entities, corridors },</code> <code>components/drill-panel/drill-panel.tsx</code> line 6: <code>import { ...</code> <code>getEvidenceForIncident,</code> <code>sampleScenarios }</code>	High Risk: Claiming real-time data, live feeds, or production analytics would be false. Data appears to be static demo fixtures for Turkey-Syria border region.
Analyst Notes Persistence	Notes are stored in component state (<code>useState</code>) only; not persisted to database or backend	<code>components/drill-panel/drill-panel.tsx</code> line 361: <code>const [notes, setNotes] = useState(...)</code> and line 365: <code>// In production, this would persist to database</code>	Medium Risk: Suggesting notes are saved to persistent storage or audit logs would be misleading.
Decision Tag Persistence	Tag changes are client-side only; no backend call	<code>components/drill-panel/drill-panel.tsx</code> line 365: <code>// In production, this would persist to database</code>	Medium Risk: Cannot claim audit-logged decision history.
Scenario Comparison Data	Sample scenarios are hardcoded in demo data	<code>components/drill-panel/drill-panel.tsx</code> line 352: <code>sampleScenarios</code> from <code>demo-data</code>	Medium Risk: Cannot claim AI-generated or real scenario analysis.
Filter Panel Time Range	Defaults to Dec 15-19, 2024; hardcoded	<code>app/page.tsx</code> line 27: <code>["2024-12-15T00:00:00Z", "2024-12-19T00:00:00Z"]</code>	Low Risk (UI/UX feature), but shows

Area	What appears mocked/demo-only	Evidence file paths	Risk if claimed publicly
			no real data integration.
Map Initial Center	Hardcoded to Idlib region (36.0, 36.6)	<code>components/map/map-canvas.tsx</code> line 120	Low Risk (intentional), but shows region-specific mock data only.
Quick Actions Buttons	"Link Evidence," "Request Imagery," "Export Report" buttons have no handlers	<code>components/drill-panel/drill-panel.tsx</code> lines 404-416	Medium Risk: Claiming functional reporting or imagery request would be false.
Admin Pages	<code>/admin</code> , <code>/models</code> , <code>/feeds</code> routes referenced in README but no component files retrieved	<code>README.md</code> lines 90-96, <code>app/admin/</code> stub	Medium Risk: May not be implemented; cannot claim admin functionality exists.
Vercel Analytics	Imported but purpose unclear; likely standard analytics only	<code>app/layout.tsx</code> line 4, <code>package.json</code> line 40	Low Risk: Standard feature, but do not claim custom intelligence metrics.

7. Technical stack

Layer	Technologies confirmed	Evidence file paths
Frontend Framework	Next.js 16.1.1, React 19.2.0, TypeScript 5	<code>package.json</code> lines 50, 52, 73; <code>README.md</code> lines 53-55
Mapping & Geospatial	Leaflet 1.9.4, React Leaflet 5.0.0, CartoDB dark basemap tiles	<code>package.json</code> lines 48, 56; <code>components/map/map-canvas.tsx</code> line 4, line 133
UI Components & Styling	Radix UI (14 component libraries), Tailwind CSS 4.1.9, Lucide React icons, next-themes	<code>package.json</code> lines 13-39, 64, 49, 51; <code>README.md</code> lines 63-66
Data Visualization	Recharts 2.15.4	<code>package.json</code> line 58; <code>README.md</code> line 60
Forms & Validation	React Hook Form 7.60.0, Zod 3.25.76, @hookform/resolvers	<code>package.json</code> lines 12, 63, 69-71; <code>README.md</code> lines 69-71

Layer	Technologies confirmed	Evidence file paths
State Management	React hooks (useState), component-level state; no Redux/Zustand/Context API visible	<code>app/page.tsx</code> , <code>components/drill-panel/drill-panel.tsx</code>
Layout & Resizing	react-resizable-panels 2.1.7	<code>package.json</code> line 57
Date & Utilities	date-fns 4.1.0, clsx, tailwind-merge, class-variance-authority	<code>package.json</code> lines 45, 43, 60, 42; <code>README.md</code> lines 74-77
Notifications	Sonner (toast notifications) 1.7.4	<code>package.json</code> line 59
Analytics	@vercel/analytics 1.3.1	<code>package.json</code> line 40; <code>app/layout.tsx</code> line 4
Build & Development	Turbopack (Next.js 16 default), pnpm 9.12.1, PostCSS 8.5, Autoprefixer	<code>package.json</code> lines 66-70, 75; <code>next.config.mjs</code> line 10; <code>README.md</code> lines 84-88
Linting	ESLint	<code>package.json</code> line 8
Database	Not found in repo.	
Backend API	Not found in repo. No <code>/api</code> routes or server actions visible.	
Authentication	No auth library found (no NextAuth, Clerk, Auth0, etc.).	
AI/LLM	No AI or LLM library found.	
File Storage	Not found in repo.	
Testing	No test framework found (no Jest, Vitest, Cypress, Playwright, etc.).	

8. Routes, pages, and user flows

Route/Page	Purpose	Evidence file paths
<code>/</code> (root)	Main workspace—interactive map with filter panel (left), drill panel (right), focus tray (bottom)	<code>app/page.tsx</code>
<code>/admin</code>	Admin interface stub (referenced but no implementation found)	<code>README.md</code> line 90, <code>app/admin/</code> directory
<code>/corridors</code>	Corridor management page (referenced but no implementation found)	<code>README.md</code> line 91, <code>app/corridors/</code> directory
<code>/entities</code>	Entity management page (referenced but no implementation found)	<code>README.md</code> line 92, <code>app/entities/</code> directory

Route/Page	Purpose	Evidence file paths
<code>/evidence</code>	Evidence management page (referenced but no implementation found)	<code>README.md</code> line 93, <code>app/evidence/</code> directory
<code>/feeds</code>	Feed ingestion page (referenced but no implementation found)	<code>README.md</code> line 94, <code>app/feeds/</code> directory
<code>/incidents</code>	Incident management page (referenced but no implementation found)	<code>README.md</code> line 95, <code>app/incidents/</code> directory
<code>/models</code>	Model management page (referenced but no implementation found)	<code>README.md</code> line 96, <code>app/models/</code> directory

Primary User Flow:

1. User loads `/` → sees map centered on Idlib region with demo incidents, entities, corridors
2. User toggles layers in left filter panel
3. User clicks on map object → drill panel opens on right with detailed view
4. User navigates drill panel tabs (Summary, Timeline, Evidence, Network, Impact, Actions)
5. User adds object to focus tray via "+" button
6. User tags object with decision tag or adds analyst note in Actions tab
7. User enters compare mode to view temporal snapshots side-by-side

9. Data model and backend logic

Data Model (from `lib/types.ts`):

- **Incidents:** `incident_id`, `type` (8 enum values: `AIRSTRIKE`, `BORDER_CLASH`, `CONVOY_ATTACK`, `IED`, etc.), `severity_score`, `confidence_score`, `status`, `geometry` (Point/LineString/Polygon), `linked_entities`, `sources` (EvidenceRef array), `analyst_notes`, `audit_log`
- **Entities:** `entity_id`, `category` (8 enum: `FACILITY`, `CORRIDOR`, `VEHICLE_FLEET`, `ARMED_GROUP`, `INFRASTRUCTURE`, `CHECKPOINT`, `MEDICAL`, `WAREHOUSE`), `criticality`, `activity_score`, `operational_status`, `linked_incidents`, `analyst_notes`, `audit_log`
- **Corridors:** `corridor_id`, `route_type` (`SUPPLY`, `EVACUATION`, `SMUGGLING`, `COMMERCIAL`, `MILITARY`), `status` (`OPEN`, `RESTRICTED`, `BLOCKED`, `CONTESTED`), `geometry` (LineString), `choke_points`, `daily_throughput_estimate`, `affected_by_incidents`

- **Evidence:** evidence_id, source_type (SATELLITE, SIGINT, HUMINT, OSINT, SENSOR, COMMS_INTERCEPT), credibility (0-1), linked_objects, collection_time, classification (UNCLASSIFIED, RESTRICTED, CONFIDENTIAL, SECRET), hash
- **OperationalImpact:** expected_delay_hours (min-max range), affected_corridors, affected_entities, cascade_risk, downstream_dependencies
- **ProvenanceBreakdown:** total_sources, sources array with credibility/weight/hours_since_collection, analyst_overrides, effective_confidence

Backend Logic:

- **None detected.** All data is client-side React components consuming hardcoded demo arrays
- No API routes (`app/api/*`) found
- No database queries or ORM visible
- No server-side business logic detected
- All filtering, selection, and state management happens in React component state

Data Flow:

- Data is imported as static arrays: `import { incidents, entities, corridors } from "@lib/demo-data"`
- Map renders these arrays directly
- Drill panel retrieves object by ID from arrays using `.find()`
- Evidence retrieved via `getEvidenceForIncident(incident_id)` helper (hardcoded lookup)
- No real-time updates, subscriptions, or polling

10. AI-specific analysis, if applicable

No confirmed AI implementation found.

- No AI/LLM libraries in dependencies (no openai, anthropic, langchain, etc.)
- No prompt engineering or LLM calls detected in code
- Confidence scores and operational impact appear to be static values in demo data
- Scenario comparison uses hardcoded `sampleScenarios` array, not AI-generated
- No evidence of inference, embeddings, or ML model integration

Confidence scoring and decision support are rule-based UI features, not AI-driven analysis.

11. Portfolio table improvements

Field	Content
Status	Prototype / Proof of Concept (v0.1.0, single commit, demo data only)
Industry	Defense / Intelligence / Geospatial Analysis
Category	B2B SaaS / Intelligence Platform / Geospatial Visualization
Project Type	Full-stack web application prototype
What we built	Interactive geospatial intelligence workstation with real-time map visualization, multi-source evidence linking, confidence scoring, scenario comparison, and decision support tools for operational intelligence analysis
Client/User Problem	Intelligence analysts need a unified workspace to visualize, correlate, and assess geospatial intelligence data (incidents, entities, corridors) with provenance tracking and confidence indicators to support operational decision-making
Main Features	Interactive Leaflet map with layer filtering, drill-down detail panels, focus tray multi-object tracking, confidence/provenance visualization, scenario comparison mode, analyst notes with audit trails, role-based access control, decision tagging, operational impact assessment
Technical Stack	Next.js 16, React 19, TypeScript, Leaflet, Radix UI, Tailwind CSS, Recharts, React Hook Form, Zod, Turbopack
Proof Available	Codebase on GitHub (private repo), visual UI/UX structure, component architecture; no production deployment, metrics, or client testimonial visible
Best Sales Angle	Fast, intuitive geospatial analysis workflow with provenance-aware confidence scoring and temporal scenario comparison for high-stakes intelligence operations
What Should NOT Be Claimed	Real-time data ingestion, live feeds, production deployment, AI/ML-powered analysis, persistent data storage, user authentication, multi-user collaboration, export/reporting functionality, backend API, or historical audit trails beyond current session

12. Safe public claims

- **Next.js and React-based geospatial intelligence workstation** for analyzing incidents, entities, and supply corridors
- **Interactive Leaflet map** with customizable layer visibility and time-range filtering

- **Multi-tab drill-down interface** for detailed object inspection including summary, timeline, evidence, network, and impact tabs
- **Confidence scoring visualization** showing source provenance and credibility breakdown
- **Focus tray** for tracking multiple objects simultaneously
- **Scenario comparison mode** with side-by-side temporal snapshot viewing and synchronized filters
- **Role-based access control** with four user roles and four classification levels
- **Analyst notes and audit trail** component for session-level decision tracking
- **Decision tagging system** (DECISION_RELEVANT, BACKGROUND_NOISE, WATCH)
- **Built with modern stack:** TypeScript, Tailwind CSS, Radix UI components, React Hook Form validation
- **Responsive, dark-themed UI** with keyboard shortcuts and terminal-style design
- **Geospatial geometry support** (Point, LineString, Polygon) for incidents, entities, and corridors

13. Unsafe or unproven claims

- **✗ Real-time data ingestion** – All data is hardcoded demo fixtures
- **✗ Live feed processing** – Feed page is a stub with no implementation
- **✗ Production-ready system** – Marked as private, v0.1.0, single commit, no deployment visible
- **✗ Persistent data storage** – No database layer; all data lost on page refresh
- **✗ Multi-user collaboration** – No authentication, session management, or backend
- **✗ AI-powered analysis** – No LLM or ML model integration
- **✗ Alert automation** – Alerts are typed but not functional
- **✗ Evidence hash verification** – Field exists but no crypto/verification logic
- **✗ Imagery request & export reporting** – Buttons exist but do nothing
- **✗ Historical audit logging** – Audit log type defined but only demos show timestamps

- **✗ Operational impact predictions** – Values are static in demo data
- **✗ Deployed/accessible system** – No deployment evidence; GitHub homepage link broken or inaccessible
- **✗ User management & security controls** – No backend auth; UI-only role definitions
- **✗ Tested/QA'd codebase** – No test files found in repo

14. Missing evidence needed

- **Screenshots** of the main map workspace, drill panel, and compare mode
- **Demo video** showing: map interaction, object selection, drill-down navigation, compare mode side-by-side workflow, focus tray usage
- **Architecture diagram** showing frontend-only vs. intended backend/database design
- **Data source documentation** – what real sources the system is intended to consume (satellite feeds, SIGINT, OSINT APIs, etc.)
- **Client/stakeholder context** – who commissioned this, what is the use case, is it a production target or prototype?
- **Performance metrics** – number of objects handled, map rendering speed, latency
- **Deployment notes** – is this intended to be deployed? Where? (GCP, AWS, on-premise, etc.)
- **Test coverage** – even basic unit/integration tests to prove functionality works
- **Production readiness checklist** – what's needed to go live? (auth, database, API, real data connectors, CI/CD, monitoring)
- **User feedback or testimonial** from an actual analyst or stakeholder
- **Case study notes** if this is tied to a real customer project or problem space

15. Final confidence score

Score: 35 / 100

Explanation:

- **Strong Codebase Fundamentals (+25 points):** Well-structured Next.js/React TypeScript project with clear component organization, proper type definitions,

responsive UI, and thoughtful UX patterns (drill panel, focus tray, compare mode).

- **Prototype-Only Limitation (~35 points):** Single commit, v0.1.0, all data hardcoded and disconnected from any real backend. No database, API, authentication, deployment, or persistence. No test coverage. Primary sub-pages (`/admin` , `/feeds` , `/incidents` , etc.) are stubs. Analyst notes and decision persistence are client-side only. Evidence hashing and alert triggering are incomplete.
- **Insufficient Portfolio Evidence (~30 points):** No screenshots, demo video, deployment link, metrics, client testimonial, or case study context. README describes intended features well but codebase implements only UI/UX layer. Cannot verify real-world use case, performance, security, or impact. Private repository with no public visibility.

Recommendation: This is a **strong design prototype** for a geospatial intelligence platform. It demonstrates solid frontend engineering and UX thinking. To strengthen as a portfolio entry, add: (1) backend mock API or real data connectors, (2) demo video walkthrough, (3) deployment to Vercel or similar, (4) case study context (who, what problem, why built), and (5) testimonial or metrics if this solved a real customer problem. If this is intended as a proof-of-concept for a client or internal tool, focus on documenting the business problem and solution impact rather than claiming production-readiness.